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A Causality-Guided Adaptive Dual-Population Evolutionary Algorithm for Constrained Multi-Objective Optimization

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***Abstract*—Most existing constrained multi-objective evolutionary algorithms primarily rely on heuristic search and empirically designed operators, often overlooking the underlying causal relationships between decision variables and objective functions. This limitation reduces their ability to avoid deceptive regions in the search space and increases the risk of premature convergence, particularly in problems with complex feasible regions. To tackle this challenge, we propose a novel causality-guided adaptive dual-population evolution for constrained multi-objective optimization algorithm (named CAD-CMOEA). The proposed CAD-CMOEA first employs an information-geometric causal inference method to model causal effects between decision variables and objectives, identifying the positive or negative influence of each variable. Based on this insight, a causality-aware genetic operator is developed by integrating competitive cooperative optimization strategies with Gaussian perturbation, enabling guided and directional variation to produce high-quality offspring. To further balance convergence and diversity, we propose a new dual-population co-evolution mechanism: one population explores high-quality solutions within the feasible region, while another population investigates promising areas near the infeasible boundary, thereby strengthening global search capabilities. Experimental results on LIR-CMOP1-14 benchmark problems demonstrate that CAD-CMOEA outperforms eight state-of-the-art algorithms in terms of IGD and HV metrics, confirming its effectiveness in causal modeling, search efficiency and solution quality.**

***Index Terms*—causal intervention, constrained multi-objective optimization, dual-population collaborative optimization, MOEA/D**

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I. INTRODUCTION

Multi-objective optimization problems (MOPs) are common in real-world applications such as engineering design [1], scheduling [2], and intelligent control [3], where multiple conflicting objectives make the solution process challenging. Multi-objective evolutionary algorithms (MOEAs) have gained attention for their ability to handle complex search spaces without relying heavily on problem-specific knowledge. In practice, many problems also involve equality and inequality constraints, leading to constrained multi-objective optimization problems (CMOPs) with increased complexity [4].

To address constraints in CMOPs, various methods have been proposed, including penalty functions [5], ϵ-constraint techniques [6], repair operations [7], and strategies that separate feasible and infeasible sub-populations [8]. While effective in certain cases, these approaches face key limitations. Most rely on heuristic rules or empirical tuning and lack the ability to extract useful knowledge from optimization data, limiting their stability and generalization in complex or dynamic environments. Additionally, in problems with sparse feasible regions or highly non-linear constraints, they often suffer from low search efficiency and are prone to local optima or stagnation.

A more fundamental limitation of mainstream constraint handling methods is their focus on numerical feasibility, while lacking the ability to model deeper dependencies among variables, constraints, and objectives. These approaches often treat the problem as a black box, emphasizing adaptive correction and feasibility guidance without exploring which variables causally affect feasibility or whether causal paths exist between objectives. Relying mainly on empirical or statistical correlations, they fail to capture the underlying structural information of complex coupled systems, limiting both search efficiency and interpretability.

Causal inference, as advanced by Pearl [9], has seen rapid progress in AI and machine learning, offering interpretable and stable models by explicitly modeling relationships between variables. It has been successfully applied in reinforcement learning [10], feature selection [11], and policy planning [12], enhancing both performance and adaptability in complex environments. However, its integration into evolutionary optimization (especially for CMOPs) remains largely unexplored. Most MOEAs still rely on heuristics and numerical search paths, overlooking causal dependencies among variables, objectives, and constraints. In real-world problems, some decision variables may directly influence constraints, while objectives may have indirect effects via mediators. Modeling these causal structures could greatly enhance search effectiveness, interpretability, and robustness, offering a new theoretical and algorithmic direction for evolutionary optimization.

Based on this, we propose a novel causality-guided adaptive dual-population evolutionary algorithm for constrained multi-objective optimization, named CAD-CMOEA. The algorithm enhances the search process by leveraging causal relationships between decision variables and objectives to efficiently explore feasible, high-quality solutions. The main contributions are:

1) Introducing information geometry causal inference (IGCI) to model causal relationships between decision variables and objectives, providing structural guidance for search direction and strategy selection.
2) Designing a genetic operator that combines causal inference with Gaussian perturbation. By identifying positive/negative influences and applying a competitive-cooperative strategy, the operator improves performance in complex search spaces.
3) Employing a dual-population collaborative framework that combines different constraint-handling methods to enhance feasible solution quality and search diversity, improving overall optimization performance in complex feasible regions.

The structure of this paper is organized as follows: Section II provides a detailed introduction to the primary mechanisms and methods proposed in this study; Section III presents the experimental results and performance analysis; Section IV summarizes the paper and proposes directions for future research.

II. PROPOSED METHOD

A. Causal Relationship Discovery Based on Information Geometry

In CMOPs, not all decision variables have a significant impact on the objective function. Failure to identify these "key variables" leads to wasted computational resources and slower convergence. To address this issue, this paper introduces Information Geometric Causal Inference (IGCI) [13], which automatically detects variables that directly influence the objective. IGCI is based on a causal inference principle: if X causes Y ($X \to Y$), the distribution $p_X(x)$ is independent of the mapping function ϕ, such that $Y = \phi(X)$; this independence typically breaks when the direction is reversed $Y \to X$. For example, if $p_X(x) = 1$ over $[0, 1]$ and f is continuous and differentiable, then:

$$P_Y(y) = \frac{1}{f'(f^{-1}(y))} \tag{1}$$

where, $P_Y(y)$ depends on $f'(x)$, highlighting the asymmetry that IGCI leverages to infer causal direction and identify key variables. By exploiting this property, IGCI evaluates whether the independence condition holds and, based on this assessment, determines the causal relationship between variables and the correct causal direction.

More specifically, if the following equality holds:

$$\int_0^1 \log \phi'(x) \cdot p_X(x)\, dx = \int_0^1 \log \phi'(x)\, dx \tag{2}$$

where, $\phi'(x)$ denotes the derivative of the function ϕ, and $p_X(x)$ represents the probability density function of the variable X. The above equality implies that there exists no statistical dependence between the distribution of X and the derivative of the function ϕ, which is consistent with the non-synergy assumption underlying the true causal direction. Accordingly, X is regarded as the causal antecedent of Y, that is, the causal direction $X \to Y$ is assumed to hold.

To further quantify the causal effect between variables and the objectives, this paper employs the Kullback–Leibler divergence (KL divergence) from information theory as a measurement tool. If X is indeed a cause of Y, then the KL divergence difference between X and Y should be less than 0. This difference is referred to as the causal effect value. The specific formula for this calculation is as follows:

$$C_{X \to Y} = D(p_X \| \mathcal{E}_X) - D(p_Y \| \mathcal{E}_Y) \tag{3}$$

where, $D(p\|q)$ denotes the KL divergence between the distribution p and the reference distribution q, and $\mathcal{E}_X$ and $\mathcal{E}_Y$ denote the families of reference distributions for X and Y, respectively (e.g. uniform or Gaussian distributions).

In implementation, a causal model is individually constructed for each decision variable dimension x_i with respect to the objective $f(\mathbf{x})$. The existence of a causal relationship is assessed using Eq.(2) and Eq.(3). Subsequently, the causal effect vectors corresponding to all decision variables in the population are extracted and utilized to guide offspring optimization in subsequent stages.

B. Causality-aware Genetic Operator

In CMOEAs, the generation of high-quality offspring is a critical factor in improving overall optimization performance. To address this, a causality-aware genetic operator is proposed in this study. The operator integrates the IGCI method [13] with the competitive swarm optimizer (CSO) strategy [14]. By quantifying the causal influence of decision variables on objective functions, the operator employs differentiated search strategies across variables, thereby facilitating the generation of more targeted and high-quality offspring. The offspring generation process consists of two primary stages: (1) CSO-guided offspring generation mechanism; (2) an adaptive perturbation guided by causal information.

In the first stage, the operator partitions the current population into a winner population (WP) and a loser population (LP) based on non-dominated sorting. Subsequently, for each individual in the LP, a winner–loser update strategy, derived from the CSO, is employed to generate preliminary offspring. This mechanism guides inferior solutions toward superior ones, thereby enhancing the overall search efficiency. The specific update equation is defined as follows:

$$\begin{aligned}\mathbf{v}_l(t+1) = r_1\mathbf{v}_l(t) + r_2\left(x_w(t) - x_l(t)\right) \\ + r_3\psi\left(\bar{x}(t), x_l(t)\right)\end{aligned} \tag{4}$$

$$x_l(t+1) = x_l(t) + \mathbf{v}_l(t+1), \tag{5}$$

where, r_1, r_2 and r_3 are random vectors sampled from the interval $(0, 1)$. The vectors $\mathbf{x}_l(t)$ and $\mathbf{x}_w(t)$ denote the corresponding solutions from the LP and WP at iteration t, respectively. The vector $\bar{\mathbf{x}}(t)$ represents the mean position of the current population. The function $\psi(\cdot)$ serves as a guidance mechanism constructed based on population-level statistical information.

In the second phase, the operator further refines the analysis of causal directionality among decision variables. It leverages the previously identified causal relationships between decision variables and objectives. Specifically, it utilizes the causal-effect vector derived during the initialization phase, along with the results of non-dominated sorting. For each decision variable, the operator calculates the difference between its values in WP and LP, assuming equal population sizes. The sign of this difference is used to determine the direction of influence. A positive value indicates a positive causal effect on the objective. A negative value suggests a negative effect. Based on this criterion, the decision variables are partitioned into two sets: positively causal variables (ind_1) and negatively causal variables (ind_2). Subsequently, the operator computes the variance of the causal-effect vector within each set. These variances are denoted as σ_1^2 for ind_1 and σ_2^2 for ind_2, respectively. A Gaussian sampling mechanism is then employed to perturb the corresponding variables. This perturbation process is used to generate a new generation of solutions. The update rule is defined as follows:

$$\begin{aligned}PopDec\{\mathrm{ind}_1\} = N(0, \sigma_1^2) * WinDec\{\mathrm{ind}_1\} \\ + LosDec\{\mathrm{ind}_1\}\end{aligned} \tag{6}$$

$$\begin{aligned}PopDec\{\mathrm{ind}_2\} = N(0, \sigma_2^2) * WinDec\{\mathrm{ind}_2\} \\ + LosDec\{\mathrm{ind}_2\}\end{aligned} \tag{7}$$

$$PopDec \in \{LosDec, WinDec\} \tag{8}$$

where, $N(0, \sigma^2)$ denotes a Gaussian distribution with zero mean and variance σ^2. WinDec and LosDec represent the decision variables of individuals in WP and LP, respectively, both having dimensions of $N/2 \times D$. This mechanism reflects an adaptive regulation of perturbation intensity based on variable sensitivity. For variables exhibiting larger variances, which indicate greater divergence in causal effects, stronger perturbations are introduced to enhance global exploration. Conversely, for variables with smaller variances, reflecting more consistent causal effects, weaker perturbations are applied to facilitate more precise local search.

C. Framework of CAD-CMOEA

Algorithm 1: Procedure of CAD-CMOEA

Input: Population size N, number of objectives M, max generations before model update $g_{\max}$
Output: Non-dominated and feasible solution set Pop_{archive}

$Pop_1 \leftarrow Initialize(N, M)$;
$Pop_2 \leftarrow Initialize(N, M)$;
$flag \leftarrow 1$;
$Pop_{\text{archive}} \leftarrow$ select N solutions from $Pop_1 \cup Pop_2$ using NSGA-II;
while *termination condition is not met* **do**
    $fr \leftarrow$ **IGCI**$(Pop_1 \cup Pop_2)$;
    $off_1 \leftarrow$ generate N offspring from Pop_1 via CI-aware operator;
    $off_2 \leftarrow$ generate N offspring from Pop_2 via CI-aware operator;
    $Pop_1 \leftarrow Pop_{\text{archive}} \cup Pop_1 \cup off_1 \cup off_2$;
    $Pop_2 \leftarrow Pop_{\text{archive}} \cup Pop_2 \cup off_1 \cup off_2$;
    $Pop_1 \leftarrow$ Select N individuals from Pop_1 using MOEA/D-Epsilon with constraint handling;
    if *Pop_2 has converged **and** $flag = 1$* **then**
        $Pop_2 \leftarrow$ Select N individuals from Pop_2 using PPS-SPEA-II with constraint handling;
        $flag \leftarrow 2$;
    else
        $Pop_2 \leftarrow$ Select N individuals from Pop_2 using PPS-SPEA-II without constraint handling;
        $flag \leftarrow 1$;
    end
    $Pop_{\text{archive}} \leftarrow$ select N solutions from $Pop_{\text{archive}} \cup Pop_1 \cup Pop_2$ using NSGA-II;
end
return Pop_{archive}

This study incorporates causal inference and causality-aware genetic operators into the CCMO framework [15], and proposes a novel optimization algorithm, called CAD-CMOEA. Algorithm 1 comprises the following three key components:

(1) Initialization: The IGCI method is employed to quantify the causal relationships between decision variables and objective value, thereby identifying the key decision

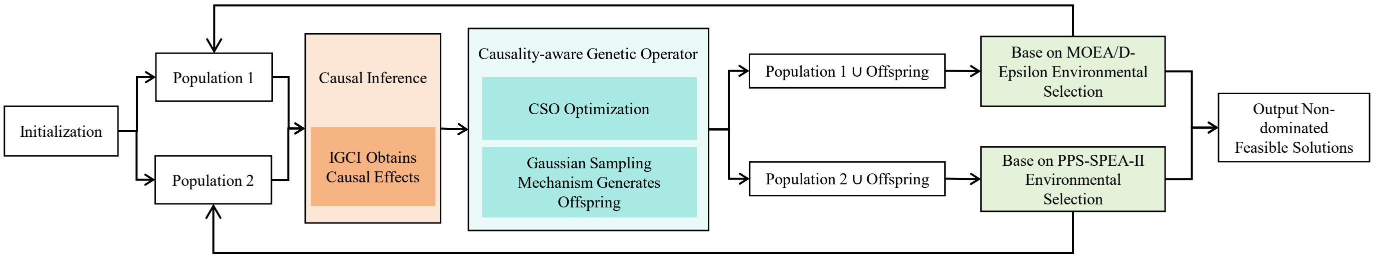

Fig. 1: Procedures of the proposed CAD-CMOEA.

variables. This step provides theoretical support for subsequent directional search, enhancing both optimization accuracy and computational efficiency.

(2) $Pop1$ Optimization Process: Upon identifying the key decision variables, offspring are generated using causality-aware genetic operators. Environmental selection is conducted using the MOEA/D-Epsilon mechanism [16], which ensures the feasibility of solutions and promotes population diversity.

(3) $Pop2$ Optimization Process: The optimization process of $Pop2$ is similar to that of $Pop1$, but there are notable differences during the search phase. The PPS [17] is integrated with SPEA-II [18] to form the PPS-SPEA-II algorithm, which is employed for environmental selection.

Algorithm 1 outlines the procedure of CAD-CMOEA. Initially, two populations, $Pop1$ and $Pop2$, are created, and IGCI is used to compute the causal-effect vector fr (Lines 1–3). A flag variable is set to 1 to indicate the search phase of $Pop2$ (Line 4). NSGA-II is then applied to the combined populations to generate a non-dominated archive Pop_archive (Line 5). In each iteration, causality-aware genetic operators generate offspring $off1$ and $off2$, which are merged with their parents to update $Pop1$ and $Pop2$ (Lines 7–10). MOEA/D-Epsilon is used in $Pop1$ to promote feasibility and diversity (Line 11). For $Pop2$, if convergence is detected and $flag = 1$, PPS-SPEA-II is used with constraint handling, and $flag$ is set to 2; otherwise, constraints are ignored to encourage exploration (Lines 13–17). Finally, NSGA-II is applied to Pop_archive, $Pop1$, and $Pop2$ to update the archive (Line 19).

To evaluate the convergence of $Pop2$, we employ a multi-metric threshold. Convergence is considered to be achieved when the rate of change in IGD and HV over the last 10 generations is below a predefined threshold, and the maximum change between the ideal and worst points does not exceed 0.001. Furthermore, convergence is enforced after 70% of the maximum iterations, even if these conditions are not fully satisfied. This ensures a smooth transition to the constrained optimization phase and guides the population toward convergence to the constrained Pareto front (CPF).

To handle constraints, we apply different constraint-handling techniques to $Pop1$ and $Pop2$. Specifically, for $Pop1$, the Epsilon-constraint mechanism dynamically adjusts constraint tightness based on the proportion of feasible solutions. It tightens the constraints when this proportion is low and relaxes them when it is high. Once 80% of the generations are completed, Epsilon is set to zero to ensure convergence toward the feasible region. For $Pop2$, constraints are initially ignored to promote progress toward the Pareto front (PF). They are then gradually introduced to guide convergence toward the CPF. This strategy balances exploration and convergence, preventing early local entrapment while improving solution quality.

III. EXPERIMENTAL STUDY

A. Experimental Setup

To evaluate CAD-CMOEA, we conducted experiments on the LIR-CMOP benchmark suite [16], comparing it with eight well-known CMOEAs: CCMO [15], CTAEA [19], EMCMO [20], NSGA-II [21], PPS [17], ToP [22], CMOES [23], MFO-SPEA2 [24]. Experimental settings are as follows:

- Decision variables: 60 for all LIR-CMOP1–14 problems.
- Population size: 100 for LIR-CMOP1–12; 105 for LIR-CMOP13–14.
- Each algorithm runs independently 30 times per problem, with 300,000 function evaluations per run.
- Comparison algorithms use parameter settings as recommended in their original papers.

B. Experimental Results

On the LIR-CMOP 1–14 benchmark suite, CAD-CMOEA and six comparative algorithms are each run 30 times. IGD results are summarized in Table I, with the best values highlighted in bold. According to the Wilcoxon rank-sum test, '+', '−', and '=' denote whether a comparing algorithm is significantly better, worse, or statistically similar to CAD-CMOEA. The results show that CAD-CMOEA outperforms most of the compared algorithms across the majority of test problems.

To evaluate the proposed algorithm, two representative problems (LIR-COMP 3 and LIR-COMP 8) are selected. LIR-COMP 3 has multiple narrow, discrete feasible regions, making CPF coverage and diversity maintenance difficult. LIR-COMP 8 contains large infeasible areas in the objective space, hindering effective convergence to the CPF.

For LIR-COMP 3, Fig. 2 illustrates the population search performance of the proposed CAD-CMOEA in comparison with eight other comparing algorithms. As shown in Fig.

TABLE I. Comparison of IGD and HV values between CAD-CMOEA and other state-of-the-art algorithms on LIR-CMOP1–14. Symbols '+', '−', and '=' indicate results significantly better, worse, or statistically similar to those of CAD-CMOEA, respectively.

Problem	Metrics	CCMO	CTAEA	EMCMO	NSGA-II	ToP	PPS	CMOES	MFO-SPEA2	CAD-CMOEA
LIR-CMOP1-14	IGD	3/11/0	2/11/1	2/11/1	0/14/0	0/14/0	1/12/1	3/9/2	0/14/0	—
(+/−/=)	HV	2/11/1	2/10/2	2/11/1	0/14/0	0/14/0	1/10/3	3/10/1	0/13/1	—

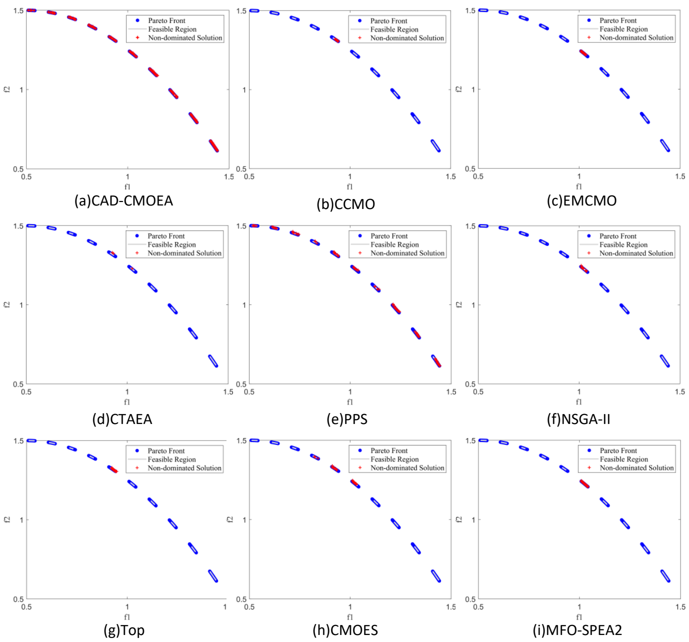

Fig. 2: The feasible non-dominated solutions with the median IGD indicator are obtained on LIR-COMP3.

2(a) and Fig. 2(e), both CAD-CMOEA and PPS successfully capture the entire CPF. CAD-CMOEA achieves this through its dual-population mechanism: one population exploits feasible high-quality solutions, while the other explores potential areas using PPS-SPEA-II, with co-evolution enhancing CPF coverage. PPS identifies the full CPF by initially ignoring constraints and converging along the UPF, laying the groundwork for later constraint handling. However, as shown in Table I, CAD-CMOEA achieves a significantly lower IGD, indicating better search accuracy and convergence on LIR-COMP 3. In contrast, the remaining algorithms fail to achieve full CPF coverage, as shown in Fig. 2(a)– Fig. 2(d) and Fig. 2(f)– Fig. 2(i). This deficiency primarily arises from their convergence-oriented search tendencies, which compromise diversity preservation. Consequently, after identifying certain segments of the CPF, these algorithms tend to become trapped in local regions due to the discontinuities among different CPF segments, ultimately failing to discover the remaining portions of the front.

For the LIR-COMP 8, Fig. 3 presents a comparative analysis of the population search performance of the proposed CAD-CMOEA and eight other benchmark algorithms. It is evident that CAD-CMOEA successfully captures the entire CPF, as shown in Fig. 3(a). This result benefits from the algorithm's key optimization designs: by incorporating causality-based variable influence information and a Gaussian perturbation mechanism, the search intensity and direction are adaptively adjusted, improving solution quality and accelerating convergence toward the Pareto front. Meanwhile, CAD-CMOEA employs a dual-population cooperative mechanism, where one population ensures solution feasibility while the other explores potential regions. Through information exchange between the two populations, the algorithm enhances search coverage and global exploration capability, ultimately enabling accurate identification of the complete CPF. In contrast, although CCMO, EMCMO, CTAEA, PPS, CMOES and MFO-SPEA2 demonstrate partial convergence toward the CPF, none are able to achieve full coverage, as illustrated in Fig. 3(b)– Fig. 3(e) and Fig. 3(h)– Fig. 3(i). This limitation primarily stems from their convergence-biased strategies in individual updating and mating selection, coupled with the absence of effective diversity preservation mechanisms. As a result, populations tend to become trapped in local regions of the front and are unable to escape from local optima, thereby constraining their ability to discover the complete CPF. The search limitations of NSGA-II and ToP are even more pronounced, as depicte in Fig. 3(f) and Fig. 3(g). Both algorithms fail to traverse several large-scale infeasible regions, ultimately resulting in incomplete coverage of the feasible front in the objective space.

IV. Conclusion

Most existing CMOEAs rely on empirical parameter tuning or heuristic rules, lacking systematic modeling of problem structures, which often leads to inefficient search and poor CPF approximation. To address this, we propose a novel causality-guided adaptive dual-population evolutionary algorithm, CAD-CMOEA. It leverages causal inference to uncover relationships between decision variables and objectives, enabling the design of causality-aware genetic operators for high-quality solution generation. A dual-population co-evolution mechanism is employed, where the primary population searches for feasible solutions and the auxiliary population explores potentially feasible regions, enhancing both convergence and diversity. Experiments on the LIR-CMOP benchmark show that CAD-CMOEA effectively captures causal dependencies and outperforms six state-of-the-art algorithms in IGD metric.

However, real-world optimization problems often involve complex and implicit causal structures, making it challenging to identify clear relationships between decision variables and objectives. To tackle this, future work will explore learning-

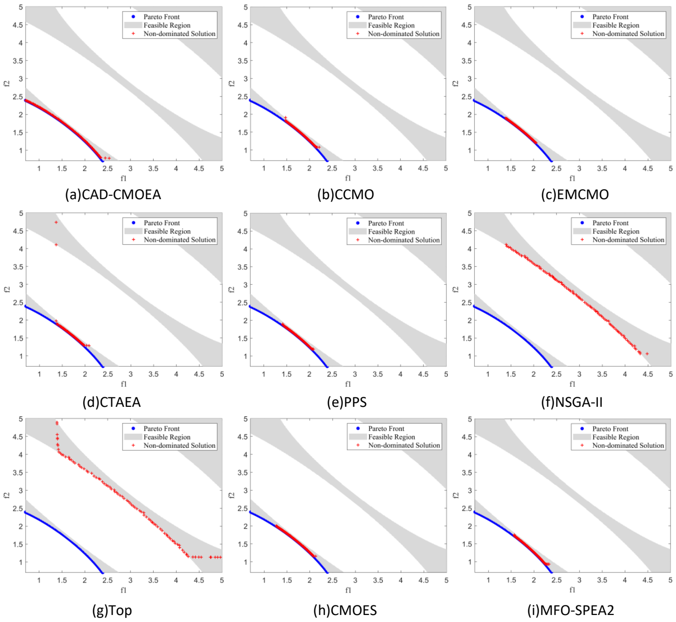

Fig. 3: The feasible non-dominated solutions with the median IGD indicator are obtained on LIR-COMP8.

based methods to embed problems into lower-dimensional spaces that facilitate causal decomposition and reveal hidden dependencies. Causal learning will also be extended to capture relationships involving constraints. Additionally, adaptive and dynamic causal regulation mechanisms will be developed to improve the accuracy and robustness of inference, ultimately enhancing search efficiency and generalization across diverse constrained optimization problems.

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